

OBSERVATIONS ON THE US MSS/GPS INTERFERENCE CONTROVERSY

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At the time of writing this column, one of the most contentious U.S. spectrum issues in recent memory is the subject of much debate. It concerns the possible use of terrestrial base station downlinks for Long Term Evolution (LTE)-based mobile communications in the Mobile Satellite Service (MSS) band just below the GPS L1 band — home of the main GPS signal for civil users. Some of the issues in this controversy are unique to the US regulatory environment while others illustrate spectrum policy issues that are more universal in nature.

This controversy arose as a result of growing demand for land mobile broadband service around the world. The FCC's National Broadband Plan [1], similar to planning in other countries, calls for 300 MHz additional spectrum in the short term and 500 MHz in the long term. Such large amounts of spectrum cannot be repurposed without impacting other services in some way. The key policy issue is what is acceptable in terms of modifying allocations for other services and changing technical conditions necessary to avoid cochannel and adjacent channel interference. While such issues have key technical components, the bottom line decision is basically a political one as the decision will inevitably result in both benefits and disruptions for various communities of users.

Intersystem radio interference has been with us ever since Marconi constructed his third radio system. A key goal of spectrum policy is to determine how tightly to pack spectrum users and what constitutes a reasonable amount of interference avoidance technology (transmitter and receiver filters, modulation details, etc.) that can be built into transmitters and receivers. It is inevitable that just as the demand for spectrum changes over time, affordable technologies to limit interference will also change. A subjective issue for spectrum regulators is thus how much interference avoidance technology to expect from transmitters and receivers, and when to change the expectations. Requiring changes too often will impose unreasonable costs on users if they must replace equipment. Being too slow results in denying spectrum to other users. Such problems have no closed form solutions based on engineering calculations.

The 1525–1559 MHz band is allocated to MSS and is adjacent to the 1559–1591 MHz GPS L1 band, which is home to both the C/A signal used for consumer GPS devices and the encrypted P(Y) or Y-code signal designed for and used by authorized military users. However, the encrypted Y-code signal is transmitted at another frequency and specialized units GPS units used in precision farming and by surveyors crosscorrelate the signals in both bands to estimate atmospheric delay and improve location accuracy in a GPS application not originally included in GPS planning.

In urbanized areas of the United States, the MSS band is very lightly used since cellular systems have achieved

near complete coverage in such areas. Multibeam satellites allow the band to be used differently in different areas. Thus, there is interest in using this MSS spectrum for terrestrial service, which is in great demand in such areas.

The basic technical problem is a classic near/far interference problem: GPS receivers are using weak signals from a constellation of satellites at an altitude of approximately 20,200 km. When located near a terrestrial base station in the MSS band, GPS receivers might experience interference from the much stronger adjacent band signal. Like *all* receivers, GPS receivers have finite front-end filtering rejection and finite dynamic range to reject strong signals from adjacent bands. The controversy regards what is reasonable for such rejection at a given point in time and the related question of what is a reasonable transition schedule. These are recurring spectrum policy questions not unique to this controversy.

In the United States, terrestrial use of the MSS band is called the Auxiliary Terrestrial Component (ATC) and has been authorized in the 1525–1559 MHz band since 2003, although there is some disagreement over exactly how many base stations were authorized at a given time. Thus, part of the contention is how much adjacent band immunity is reasonable for GPS equipment built after the 2003 policy change. Another question is the specific receiver design used by some manufacturers of precision GPS equipment. This precision equipment uses both the C/A and P(Y) GPS signals as well as an augmentation signal in another band that enables more precise positioning using the Real Time Kinematic [2] algorithm. In the controversial receiver design here, certain RTK systems uses one broadband front end that covers the MSS band (which contains an Inmarsat downlink sometimes used for the augmentation signal), the GPS band, and the band used by the Russian Glonass GPS-like system, with total front end coverage from 1525 to 1610 MHz. Such a broadband receiver is very vulnerable to near/far problems since *any* strong signal within its bandwidth is likely to exceed its dynamic range limitations. Whether such receiver design is reasonable after the initial 2003 authorization of terrestrial ATC transmitters in the MSS band is one of the points of contention in this issue.

The U.S.-specific aspects of this problem deal with the split spectrum management responsibilities in the unique U.S. regulatory structure. National government (federal) spectrum users are regulated by the National Telecommunications and Information Administration (NTIA), while private sector and state and local government users are regulated by the Federal Communications Commission (FCC). The two agencies have agreed to coordinate with each other decisions that would impact each other's jurisdiction [3]. Thus, the GPS satellites and many of the GPS users are authorized by NTIA, while any use of the MSS band is regulated by FCC. NTIA generally believes in

receiver regulation as a key part of spectrum management and has jurisdiction to regulate receivers of federal users. However, FCC has little authority to regulate receivers and generally has declined to use some of the limited authority [4] it has. Thus, GPS receivers outside the federal government are subject to no mandatory interference rejection standard.

Complicating the issue is the fact that two other government entities, the Federal Aviation Administration (FAA) and the Department of Defense (DoD), are chafing under the current regulatory system and past and pending reallocations of spectrum to FCC-regulated commercial users. In 2006, FAA even tried to unilaterally impose itself as a coequal spectrum regulator [5] by proposing to veto assignments/licenses made by FCC or NTIA. However, FAA withdrew this proposal under widespread criticism. FAA and DoD have been very supportive of GPS industry concerns in this controversy, and have had an active role in testing available GPS equipment and have generally been silent on what interference rejection goals are reasonable for GPS equipment now and in the future.

An old saying in the spectrum policy field is “receivers use spectrum, not transmitters.” That is, most of the spacing between transmitter frequencies and the spatial reuse of frequencies is often governed not by the necessary bandwidth of the modulations used, but by the limitations of practical receivers and the near/far ratios likely to be encountered in the geometry of actual usage. As technology, particularly filter technology, improves, it becomes possible to use spectrum more intensively if receivers are more robust with respect to undesired signals. However,

this is an “externality” in economics jargon: the entity that incurs the cost (both engineering cost and marginal production cost) of better receivers does not necessarily benefit directly from the more intense spectrum use that is enabled. Indeed, if use of improved receiver technology obsolesces equipment before its planned lifetime, the entity owning the receiver may have considerable costs imposed on it. This is why spectrum policy is not purely a technical area: public interest factors have to be balanced with technical factors in order to determine whether costs incurred are comparable to public interest benefits and whether costs are distributed equitably.

REFERENCES

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BIOGRAPHY

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